7.5.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the exploration and activity that I knew were …	Something I wondered during the exploration or activity was …	Something I learned during the exploration or activity was …

Unit 7, Lesson 6: Factoring by Grouping

Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.

Learning Target

- ☐ I can factor a polynomial by grouping.

7.6.1 Warm-Up: Notice and Wonder

1. Coral Castle in Miami, Florida was created by Edward Leedskalnin. He carved over 1100 tons of coral rock using an unknown process, often carving at night.

   a. What do you notice about the sculptures?

   b. What do you wonder about the process Ed used?

7.6.2 Guided Instruction: Factoring by Grouping

When factoring $ax^2 + bx + c$, the coefficient b is the sum of factors of the product of a and c.

1. In both $8x^2 - 19x - 15$ and $8x^2 - 2x - 15$, the product of a and c is -120.

   a. List all the pairs of factors for -120.

   b. Circle the factor pair for -120 that adds to the sum of -19.

   c. Draw a rectangle around the factor pair for -120 that adds to the sum of -2.

2. The factor pairs can be used to factor by grouping. Complete the steps for factoring by grouping for each trinomial.

$8x^2 - 19x - 15$	$8x^2 - 2x - 15$
$8x^2 - 24x + 5x - 15$	$8x^2 - 12x + 10x - 15$
$8x(x - \_\_\_\_\_) + 5(x - \_\_\_\_\_)$	$4x(2x - \_\_\_\_\_) + 5(2x - \_\_\_\_\_)$

Factoring by grouping is helpful when factoring trinomials of the form $ax^2 + bx + c$, where the value of a has multiple factors.

3. Factor $6x^2 - 19x + 15$ by grouping.

Factoring by grouping can also be helpful when factoring a polynomial with four terms.

4. Factor $3y^4 - 6y^3 + 9y^2 - 18y$ by grouping.

5. The polynomial $3y^4 + 9y^2 - 6y^3 - 18y$ is a rearranged form of $3y^4 - 6y^3 + 9y^2 - 18y$.

   Factor $3y^4 + 9y^2 - 6y^3 - 18y$ by grouping.

6. What do you notice about the factored forms of $3y^4 - 6y^3 + 9y^2 - 18y$ and $3y^4 + 9y^2 - 6y^3 - 18y$?

7.6.3 Guided Practice: Factoring by Grouping

1. Factor $6x^2 + x - 35$ by grouping.

2. Factor $-15g + 35gh - 7h + 3$ by grouping.

7.6.4 Collaboration: Factoring by Grouping

Work with your partner to complete the following.

1. Factor $2x^2 + 3x + 4x + 12$ by grouping.

2. What happens when you try to group the expression and find a common factor?

3. If you rearrange the middle two terms, what happens when you try to factor by grouping again?

4. Complete the statement.

   When factoring an expression by grouping, group terms

   that have coefficients that
   - ○ are exactly the same.
   - ○ have common factors.
   - ○ are in numerical order.

5. Work with your partner to create a strategy for the best way to group an expression.

7.6.5 Your Turn!

1. Factor $12x^2 - 16x - 3$ by grouping.

7.6.6 Wrap-Up: Ticket Out the Door

1. Factor each expression.

   a. $h^2 - 12h + 27$

   b. $36a^4b^7 + 42a^6b^3$

   c. $12x^2 - 8x - 15$

   d. $5x^2 + 37x + 14$